**Youth Skill Development Training Program**

**Batch: Nov’24 ~ Feb’25**

**Final Project for Odoo Team**

# 1.Project Overview:

The project aims to design and implement a system that facilitates supplier registration, RFP creation, and quotation management. The solution should streamline supplier account creation, RFP publishing, and the evaluation process, while offering insightful reporting and dashboard features for better decision-making.

# 2.Core Features:

| Feature ID | Feature Summary |
|---|---|
| 1 | Email sign-up and OTP validation |
| 2 | Supplier registration form submission |
| 3 | Two-Step Verification Process on Supplier Account Creation |
| 4 | RFP creation, publication & quotation submission |
| 5 | Report preparation |
| 6 | Dashboard preparation |
| 7 | Prepare Deliverables |

# 3.Users Activity:

| User Group | User Type | Activity | Security |
|---|---|---|---|
| Supplier | Portal | Email verification using OTP<br>Supplier Registration Form.<br>View Own Profile.<br>View RFP<br>Submit Quotation against RFP | Own records only |
| Reviewer | Internal | Create a RFP (Draft, Submit)<br>Review Supplier Registration<br>Form (Recommend / Reject / Blacklist, Feedback)<br>Review Quotations from | Own records only |

| | | | |
|---|---|---|---|
| | | supplier (Recommend / Score) | |



| | | | |
|---|---|---|---|
| Approver | Internal | Review and Approve / Reject Supplier Registration.<br>Review and Approve / Publish RFP, Reject<br>Review Publish RFP | All records |

# 4.Feature wise Requirement Details:

## Feature 1: Email Verification with Evaluation

- **Email Input:** User enters an email address.
- **Evaluation Checks:** Verify if an account is already associated with email.
- **Blacklist Chec**k: Verify if the email is blacklisted.
- **Alert/Notification:** If the email is blacklisted or If the email is already registered.
- **Send OTP:** If the email passes both checks, send an OTP to the email address.
- **Alert the user:** *"OTP has been sent to your email."*
- **OTP Verification:** Provide an input field for the user to enter the OTP.
- **Validate the OTP:** within the time limit. If OTP is invalid, alert the user: *"Invalid OTP. Please try again."*
- **If the OTP is valid**: proceed to the registration form. Alert the user: *"OTP verified. Proceeding to registration."*

## Feature 2: Registration Form Design with Vendor Integration

Objective: Design a registration form divided into 5 sections, ensuring the reuse of fields already defined in the Vendor model. Add new fields only when necessary and modify the vendor form, list and other views to include these fields. You can use notebook pages and a Kanban view for better visibility.

- Section 1: **Supplier Company Information and contact person information**
- Section 2: **Financial Information**
- Section 3: **Previous Client References**
- Section 4: **Certification**
- Section 5: **Document Submission**

| |
|---|
| **Section 1: Company Information** |

| | |
|---|---|
| <u>**Mandatory Fields**</u><br>**Company Name**: Must be provided. **Company Address**: Must be provided. **Company Type**: Dropdown (e.g., LLC, Corporation). | <u>**Optional Fields with Validation**</u><br>1. **Trade License Number**: Optional but validated:<br>○ Must be alphanumeric, 8–20 characters. |



| **Section 1: Company Information** | |
|---|---|
| **Company logo:** (Include fields for the company logo) | 2. **Commencement Date**: Optional but validated:<br>○ There must be a date **before today** if provided.<br>3. **Expiry Date**: Optional but validated: ○ Must be a **future date** if provided; alert if expired.<br>4. **Tax Identification Number (TIN)**: Optional but validated:<br>○ Must be a **16-digit numeric string** (e.g., "1234567890123456"). |
| <u>**Contact Information:**</u><br>**Primary Contact**: Name, Email, Phone, Address<br>**Finance Contact:** Name, Email, Phone, Address<br>**Authorized Contact**: Name, Email, Phone, Address. | |

| **Section 2: Bank Information** |
|---|
| **Bank Name:** Input field for the name of the bank where the company holds an account. (Mandatory)<br>**Bank Address:** Input field for the complete address of the bank branch managing the account. (Mandatory)<br>**Bank SWIFT Code:** Input field for the bank's SWIFT code, which is essential for international transactions<br>**Account Name:** Input field for the name under which the bank account is registered, typically the company name.<br>**Account Number:** (Input field for the bank account number. (Mandatory)<br>**IBAN:** Input field for the International Bank Account Number, used for cross-border transactions |

### Section 3: Previous Client References Form

**Client 1:** Name, Email, Phone, Address
**Client 2: Name**, Email, Phone, Address
**Client 3: Name**, Email, Phone, Address
Facilitate adding up to 5 client references. If any Email, Phone, Address is provided then name is mandatory.

### Section 4: Certification



**Certification:** Input field for the name or title of the certification or accreditation **Certificate Number:** Input field for the unique identifier of the certification

**Certifying Body:** Input field for the name of the organization or authority that issued the certification

**Award Date:** Date picker for the date on which the certification was awarded **Expiry Date:** Date picker for the expiration date of the certification, if applicable, expired certification cannot be added

### Section 5: Document Submission

- Trade License / Business Registration
- Certificate of Incorporation
- Certificate of Good Standing
- Establishment Card
- VAT / Tax Certificate
- Memorandum of Association / Power of Attorney / Legal Instrument authorizing Authorized Person
- Identification Document for Authorized Person / Signatory
- Bank Letter indicating Bank Account Information
- Past 2 years of Audited Financial Statements
- Other Certification / Accreditations (Specify any additional documents not listed above)

**Note that file size cannot exceed 1 MB**

**Declaration:** Before submitting your documents, please read and confirm the following statement:
"I confirm that all information and documents supplied are true and accurate to the best of my knowledge, and I understand that this information may be considered material in the evaluation of quotations, bids, and/or proposals."

**Name of Signatory:** (Input field for the name of the person confirming the information, Mandatory) **Authorized Signatory:** (Input field for the title or role of the signatory within the company, mandatory)
**Company Stamp & Date:** (Include fields for the company stamp and the date of submission, mandatory)

Upon submission, the system should automatically validate the application form for any missing or invalid data.
If any issues are detected, notify the user with clear instructions on what needs to be corrected before resubmission.



If submission is successful, display a success message confirming receipt of the application.
Notify reviewers to review this application via email.
All assigned reviewers should receive email notification prompting them to review the newly submitted application.

## Feature 3: Two-Step Verification Process on Supplier Account Creation

1. First Review (User of reviewer group)
   - o Reviewer's Action:
     - o The first reviewer examines the application details and can choose to "Approve," "Reject," or "Blacklist", the applicant.
     - o If "Reject" or "Blacklist" is selected, provide an option to include comments explaining the decision.
   - o Next Steps:
     - o Upon approval, the application is forwarded to the approver for final review.

2. Final Approval:

o Approver's Action:

● The approver reviews the first reviewer's decision and any additional comments. ● The approver provides the final decision, either approving the creation of a vendor record or rejecting the application.

3. Vendor Creation:

o Upon final approval, automatically create a vendor record in the system along with a user account for the supplier.

4. Supplier Notification:

o Inform the supplier via email about the successful creation of their vendor record. o Include login credentials in the email for accessing their account in the supplier portal.

## 5.UI Views

To ensure a smooth user interface and user experience, the following views should be developed:

● **From View:** Use a clean and organized form layout to collect all necessary supplier data. Ensure responsive design for various devices.

● **Tree View:** Provide a summarized view of ongoing applications, displaying key details such as applicant name, status, and dates for easy tracking.



● **Kanban View:** Utilize a kanban board to visually manage the application process. Organize applications into columns such as "Under Review," "Approved," and "Rejected."

● **Search and Filter:** Implement robust search and filtering capabilities to allow reviewers to quickly find and prioritize applications by various criteria, such as date, status, or applicant name.

**Implementation Note:** Ensure all views support real-time updates and transitions between stages of review, facilitating efficient workflow management. Notifications should integrate seamlessly with email systems and provide direct links to the application details for ease of access by reviewers. Consider security measures for sensitive supplier data, especially in handling login credentials.

## Feature 4: RFP creation, publication & quotation submission

RFP means Request for Purchase. This RFP should be created by users of the Reviewer group only (i.e. reviewer) and each reviewer can see only their own RFP.

RFP should have a **list** and **form** view both in the **portal** and **backend views**. You need to keep track of the field changes in the chatter log of backend form views. Also 2 notebook pages, one for the product lines and another for the RFQ (Request for Quotation) lines. Below will be features:

- **RFP Menu in Portal View:** A menu named **RFP** is added to the portal. Clicking displays a list view of all RFPs.
- **List View of RFPs:** Shows all RFPs with filtering, searching, and grouping options.
- **Form View for RFP Details:** Clicking an RFP opens a form view where suppliers can:
    - o Enter product prices, quantities, and delivery charges.
    - o Submit their quotations.
- **RFQ Creation:** Upon quotation submission:
    - o An RFQ is created with supplier data, pricing, and delivery details.
    - o The RFQ is linked to the corresponding RFP.
- **Email Notification:** An email is sent to the reviewer upon quotation submission.
- **Quotation Review:**
    - o Reviewers can evaluate quotations from backend views.
    - o Score based on delivery date, charges, warranty, and terms.
    - o Recommend preferred suppliers.

RFP should contain the following fields listed below:

1. RFP number: Char. Sequence number the system generates when the record is created.
2. Status: Draft, Submit, Approved, Rejected, Closed, Recommendation, Accepted. Default value: Draft
3. Required date: Date field. Default: 7 days later from the current date.
4. Total Amount: Computed and stored from costs in the accepted RFQ line.

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5. Log access fields

6. Approved Supplier: Many2one. Editable by approver only. And domain is recommended supplier only.
7. Product lines: One2many field for reviewer containing the following info:-
    - Product: Many2one
    - Description: Text
    - Quantity: Integer
    - Unit price: monetary (not shown in tab of RFP form)
    - Subtotal price: monetary (not shown in tab of RFP form)

        - Delivery Charges: monetary (not shown in tab of RFP form)
8. RFQ lines: One2many fields using purchase. Order model. Must display the following fields from RFP:
    - Supplier: Many2one
    - Expected delivery date: Date
    - Terms and Conditions: HTML/Text
    - Warranty period in months: Integer
    - Score: Integer
    - Recommended: Boolean. **Note that a supplier cannot have more than one recommended RFQ line.**
    - Product lines: One2Many. Separate data generated by supplier from portal and only shows on inline form view of field.
    - Total price: Computed from (quantity * unit price) + delivery charge of its product lines
9. Actions:
    a. Reviewer's actions:
    - **Button Name: Submit**
        - Visibility: After record has been created and in Draft
        - Action:
            - When in submitted status, the form cannot be edited
            - Listed in the approver's menu for review.
            - When the action is performed, it will also notify the approver
    - **Button Name: Return to Draft**
        - Visibility: when status is submitted.
        - Action: Return status to draft for making necessary changes.
    - **Button Name: Recommend**
        - Visibility: When the record is in a closed state
        - Action:
            - Listed in approver's final review list.
            - Notify the approvers via email



            - Raise exception if none of the RFQ lines has been recommended

    b. Approved actions:
    - **Button name: Approve**

○ Visibility: when status is submitted.
○ Action:
■ Notify reviewer about the reviewer of the approval via email
■ Notify suppliers of newly added RFP and show this RFP as a list item from the portal menu Submit Quotation
- **Button name: Reject**
○ Visibility: when status is submitted.
○ Action: Notify reviewer about the rejection via email
- **Button name: Close**
○ Visibility: when status is Approved.
○ Action: Remove RFP from the portal
- **Button name: Accept**
○ Visibility: when status is Recommendation.
○ Action: Create PO from the **Approved Supplier** using their corresponding RFQ line.

## Feature 5: Report on RFP

Create a menu for generating RFP reports. The menu will only show a form view, and the report model uses the data from its fields momentarily to generate reports based on a button click. Therefore, it must not create new records and contain 2 buttons, each with a separate function:

1. Creating a Qweb-HTML preview
2. Generating an Excel report using the xlsxwriter package

The report model contains the following fields:

1. Supplier: Many2one. Required. It is used for finding RFPs based on suppliers of the current company.
2. Start date: Date. Required. The required date field of RFPs must be >= this model's start date.
3. End date: Date. Required. The required date of RFPs must be <= this model's end date. The Excel report must include the following specifications:

- The start date must be <= the end date; otherwise, it will throw an exception.
- Check if the given supplier has any accepted RFPs. If not, it must throw an exception.
- In Section-1:
  ○ Place the logo of the current company at the top left of the report. You can retrieve information on the current company in any appropriate way and if the current company doesn't have a logo, you must throw an exception for adding it.

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○ Some columns after the logo display the following supplier/vendor information: ■ In a title style, display the supplier’s name.

■ Below it, there should be a table containing other supplier information: email, phone, address, TIN and its related bank name, account name, account number, IBAN no., and SWIFT code.

An illustration is given below as an example:

- In Section-2, display a table containing RFPs for which the supplier is an approved supplier. The table should show columns for the RFP number, date, required date, and total amount of an RFP. The last row of the table will show the net amount of all RFPs. The dates must be in dd/mm/yyyy format.
- In Section-3, display a table showing product lines of different accepted RFPs of the given supplier. The product lines should be grouped by product to display the columns for the product name and summation results of the following fields: quantity, unit price, delivery charge, and subtotal price. The last row of the table will show the total price calculated from all product lines and should be the same as the net amount displayed in the table of Section-2.
- In Section-4, display the current company's email, phone number, and address on separate lines on the left side.

The HTML preview will look similar to the Excel report except for the fact that the header will only show the company logo with a horizontal line below, and the supplier information is displayed beneath the horizontal line:

## Feature 6: Data Visualization

RFP data should be visually represented for analysis using the following views:

- **Graph view**

1. By default, it should display a line graph with required date as x-axis and total amount as y-axis.
2. The x-axis must show data points on a monthly-basis
3. The legend must show RFP number, supplier, and status
4. Data displayed by bar and line graphs must initially be in an unpacked state.
5. Disable redirection from click on graph view.

- **Pivot view**
  1. Supplier will be the main row
  2. Required Date will be the main column and be displayed month-wise.
  3. There will be a Status sub-row
  4. Data to be measured and displayed together by default are count and total amount.
  5. Measured data should be displayed in descending order.

### Feature 7: Dashboard

Develop a dynamic and interactive dashboard that allows users to select a supplier from a dropdown list and specify a date range (e.g., this week, last week, last month, last year, etc.). The dashboard will then display key metrics for the selected supplier within the chosen time period. These metrics should include:

1. **Approved RFQs:** Display the total number of approved Requests for Quotation (RFQs) for the selected supplier within the specified date range.
2. **Total Amount:** Show the total value of approved RFQs for the selected supplier during the specified date range.
3. **Product-Wise Breakdown:** Include a product-wise count board that displays the name and quantity of each product related to the selected supplier within the given period.

# 6.Innovation Opportunity

In addition to the core functionality, trainees are encouraged to enhance the dashboard by incorporating relevant graphical representations, such as bar charts, pie charts, or line graphs, to visually present the data and provide deeper insights.

# 7.Technical Requirement

The dashboard must be developed using the OWL component to ensure consistency with the project's technical standards and framework.

### 1. OWL (Odoo Web Library) Integration

- **Requirement:** The dashboard must be developed using the OWL (Odoo Web Library) component to ensure consistency with the project's technical standards and framework.

- **Clarification:** Ensure the use of OWL components for UI rendering and state management, following the Odoo framework's architecture and best practices.

### 2. Performance Optimization

- **Requirement:** The dashboard should be optimized for performance, ensuring fast loading times and efficient rendering of components.
- **Clarification:** Use OWL's reactive state management effectively to minimize unnecessary re-renders. Lazy loading or on-demand loading of data should be implemented where applicable.

### 3. Responsiveness and Mobile Compatibility

- **Requirement:** The dashboard must be responsive and compatible with a variety of screen sizes, ensuring smooth user experiences across desktop and mobile devices.
- **Clarification:** Implement responsive design principles using OWL components and CSS frameworks (like Bootstrap or Odoo's own CSS utilities) to handle different screen sizes effectively.

### 4. Data Binding and State Management

- **Requirement:** The dashboard should support efficient data binding and state management using OWL’s built-in features.
- **Clarification:** Ensure that OWL’s reactive state management (using `useState` and `useEffect`) is leveraged for maintaining and updating UI components based on dynamic data.

### 5. User Authentication and Access Control

- **Requirement:** The dashboard must respect Odoo's existing user roles and permissions for data access control.
- **Clarification:** Implement access control logic to show or hide specific components based on user roles, utilizing Odoo's `ir.model.access` and group rules in conjunction with the dashboard components.

### 6. Error Handling and Logging

- **Requirement:** Proper error handling and logging should be implemented to ensure any issues are easily traceable and resolved.

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- **Clarification:** Utilize Odoo’s error-handling mechanisms and logging features to capture and display error messages or warnings where necessary.

### 7. Data Visualization

- **Requirement:** The dashboard should support effective data visualization (charts, graphs, tables, etc.).
- **Clarification:** Use compatible libraries for rendering visual components (such as `Chart.js`, `Plotly`, or `Odoo's built-in graphing tools) in combination with OWL components for smooth integration.

### 8. UI/UX Consistency

- **Requirement:** The dashboard must adhere to Odoo’s UI/UX guidelines for consistency across the platform.
- **Clarification:** The design and interactions should align with Odoo’s general look-and-feel, using Odoo’s UI components and styles where possible.

### 9. Security and Data Integrity

- **Requirement:** Ensure secure handling of sensitive data displayed or processed by the dashboard.
- **Clarification:** Adhere to security best practices, such as using Odoo’s ORM methods to access data (ensuring SQL injection prevention), validating user inputs, and encrypting sensitive data when necessary.

### 10. Testing and Quality Assurance

- **Requirement:** The dashboard should be thoroughly tested to ensure functionality, performance, and security.

  **Clarification:** Implement unit tests for OWL components, as well as integration tests where appropriate, to ensure proper behavior under different scenarios.

# 8.**Guidelines:**

For most cases, you should follow the coding guidelines specified in the following link:
[https://www.odoo.com/documentation/17.0/contributing/development/coding_guidelines.html](https://www.odoo.com/documentation/17.0/contributing/development/coding_guidelines.html)
Since it will be a learning project as part of Academic training completion, using any AI assistant tool to implement any features or module will be strictly prohibited.

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# 9.**Deliverables**:

## 9.1 Docker Containerized Application:

o The entire application must be containerized using **Docker**.
o The application should run smoothly within the container, encapsulating all necessary dependencies and configurations for the system.
o The Docker container must be properly configured to ensure that all required services (e.g., web server, database, etc.) are running within the container.

## 9.2 Nginx Configuration:

○ The application will be served behind Nginx as the web server, acting as a reverse proxy to handle requests and route them to the application backend.
○ Proper configuration of **Nginx** should be included to ensure smooth handling of traffic, including features like load balancing, SSL termination (if applicable), and URL routing.

## 9.3 Proper Documentation:

○ **Dependence Documentation:**
■ A clear list of all the dependencies required for the application (e.g., libraries, frameworks, databases, etc.) and how they interact with each other.
■ Any specific version requirements and compatibility notes should be included. ○ **Process Installation Documentation:**
■ Step-by-step guide on how to install and configure the application within the Docker container.

■ Instructions on how to build the Docker container, run it, and deploy it in different environments (e.g., local machine, production server).
■ Detailed steps on how to set up **Nginx** and integrate it with the Dockerized application.
■ Information on how to troubleshoot common installation or configuration issues.